

Special Relativity

Labix

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1 Mechanics in Special Relativity

1.1 Minkowski Spacetime

Einstein's two postulates.

Postulate: Homogeneity of space and time and isotropy.

From the above, conclude that spacetime is Minkowski.

Definition 1.1.1 (Minkowski Spacetime)

Define the Minkowski spacetime by the following data.

- A four dimensional affine space $(M, V, +)$.
- A non-degenerate symmetric bilinear form $\eta : V \times V \rightarrow \mathbb{R}$ of signature $(3, 1)$.

By Sylvester's law of inertia, we are guaranteed global Lorentz orthonormal bases $E = \{v_0, \dots, v_3\}$. These are the bases for which

$$[\eta]_{E,E} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Typically, we think of η has having the matrix

$$\begin{pmatrix} -c^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where c is the speed of time. But the geometry does not differ other than by a scalar because it only depends on the signature.

We recall some facts about affine spaces that makes them a smooth manifold.

- For each $p \in M$, the map $\phi_p : M \rightarrow V$ given by $\phi_p(x) = x - p$ does not define a global chart on M . We need to choose a basis for V to get an isomorphism $V \cong \mathbb{R}^4$ for the global chart.
- For each $p \in M$, the map $\alpha_p : T_p M \rightarrow V$ given by $[\gamma] \mapsto \left. \frac{d}{dt} \right|_{t=0} (\gamma(t) - p)$ is a canonical isomorphism. This means that no connection is needed to compare the fibers of the tangent bundle. The structure of the affine space means inherently that M is flat.

We also recall some definitions of Lorentzian geometry. Let V be a vector space of dimension 4. Let η be a non-degenerate symmetric bilinear form on V of signature $(3, 1)$. Let $v = (t, v_1, v_2, v_3) \in \mathbb{R}^4$.

- We say that v is space-like if $|v|_\eta > 0$.
- We say that v is time-like if $|v|_\eta \in \mathbb{C} \setminus \mathbb{R}$.
- We say that v is light-like if $|v|_\eta = 0$.

Definition 1.1.2 (Time-Orientation)

Let $(M, V, +, \eta)$ be Minkowski spacetime. A time orientation of M is a choice of connected component of the set of time-like vectors $\{v \in V \mid |v|_\eta \in \mathbb{C} \setminus \mathbb{R}\}$.

Definition 1.1.3 (Future Pointing Vectors)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Fix a time orientation of M . Let $v \in V$ be a time-like vector. We say that v is future pointing if v lies in the connected component defining the time orientation.

Definition 1.1.4 (The Minkowski Metric)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Define the Minkowski metric $\eta \in \mathcal{C}^\infty(T^*M \otimes T^*M)$ as

follows. For each $p \in M$, define

$$\eta_p(v, w) = \eta(\alpha_p(v), \alpha_p(w))$$

where α_p is the isomorphism $\alpha_p : V \xrightarrow{\cong} T_p M$.

The Lorentzian properties on V translate canonically to $T_p M$ via α_p . This gives M the structure of a pseudo-Riemannian manifold.

Proposition 1.1.5

Let $(M, V, +, \eta)$ be Minkowski spacetime. Then (M, η) is a pseudo-Riemannian manifold with metric signature $(1, 3)$.

For a curve $\gamma : \mathbb{R} \rightarrow M$ in Minkowski space time, the vectors $\gamma'(t) \in T_p M$ is either space-like, time-like or light-like. Solving the equation $|x|_\eta = 0$ gives a 3-dimensional cone in $\mathbb{R}^4$.

- If $\gamma'(t)$ lies on the light cone, then $\gamma(t)$ is travelling at the speed of light.
- If $\gamma'(t)$ lies inside the light cone (time-like), then this is physically realizable as an object travelling through space time.
- If $\gamma'(t)$ lies outside the light cone (space-like), then it is physically impossible.

Definition 1.1.6 (Inertial Frame of Reference)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Fix a time orientation of M . An inertial frame of reference is a chart $(M, \phi_E \circ \phi_p)$ where

- $p \in M$ is a choice of origin.
- $E = \{e_0, \dots, e_3\}$ is a Lorentz orthonormal basis of V such that e_0 is future pointing.
- ϕ_E is the induced isomorphism $\phi_E : V \xrightarrow{\cong} \mathbb{R}^4$.

Subtle: since all tangent spaces are canonically V , fixing an orthonormal basis means that the frame is of constant speed.

Definition 1.1.7 (The Poincare Group)

Let V be a vector space of dimension 4. Define the Poincare group to be

$$P = O(1, 3) \ltimes V$$

Proposition 1.1.8

Let $(M, V, +, \eta)$ be Minkowski spacetime. Then there is an isomorphism

$$\text{Isom}(M, \eta) \cong O(1, 3) \ltimes V$$

Proposition 1.1.9

Let $(M, V, +, \eta)$ be Minkowski spacetime. Then the Poincare group is the largest subgroup of $\text{Aff}(V)$ that maps inertial frames to inertial frames. Also it acts transitively on inertial frames.

Definition 1.1.10 (Translations, Boosts and Rotations)

Let $T \in O(1, 3) \ltimes \mathbb{R}^4$.

- We say that T is a boost if T is of the form

$$T(t, x) = \left(\frac{t - v \cdot x}{\sqrt{1 - v^2}}, \frac{\langle x, v \rangle - |v|t}{\sqrt{1 - v^2}} + \left(x - \frac{\langle x, v \rangle}{|v|^2} v \right) \right)$$

for some $v \in \mathbb{R}^3$ such that $|v| < 1$, and $x = \frac{\langle x, v \rangle}{|v|^2} v + \left(x - \frac{\langle x, v \rangle}{|v|^2} v \right)$ is the orthogonal decomposition.

- We say that T is a translation if T is of the form

$$T(t, x) = (t + t_0, x + x_0)$$

for some $x_0 \in \mathbb{R}^3$ and $t_0 \in \mathbb{R}$.

- We say that T is a rotation if T is of the form

$$T(t, x) = (t, Gx)$$

for some $G \in SO(3) \leq SO^+(1, 3)$.

Boosts are different in Minkowski spacetime because we assume $c = 1$ and no velocity can reach $c = 1$. So $|v| < 1$.

In the Galilean spacetime these three types decomposes the Galilean group. In Minkowski spacetime, boosts and rotations are given by the Cartan decomposition $SO^+(1, 3) \cong \{\text{Boosts}\} \times SO(3)$ as a smooth manifold (not as a group). Combining the translations give the restricted Poincare group, which is not the full Poincare group.

Definition 1.1.11 (The Restricted Poincare Group)

Define the restricted Poincare group to be

$$P^+ = \mathbb{R}^4 \rtimes SO^+(1, 3)$$

Proposition 1.1.12

There is an isomorphism

$$P \cong P^+ \ltimes_{\phi} (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$$

where $\phi : \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \rightarrow \text{Aut}(P^+)$ is given by:

1.2 Relativistic Kinematics

Definition 1.2.1 (World Lines)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a curve. We say that r is a world line if $r'(t)$ is future pointing for all $t \in \mathbb{R}$.

Physically: time is coordinate dependent, proper time is the absolute notion that everyone agrees on.

Definition 1.2.2 (Proper Time)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line. Let $t_0 \in I$. Define the proper time of r to be

$$\tau(t) = \int_{t_0}^t \sqrt{-\eta(r'(s), r'(s))} ds$$

for $t \in I$ and $t > t_0$.

Heuristically, we have

$$d\tau = \sqrt{-\eta(r'(s), r'(s))} dt$$

The choice of coordinate frame changes how η and dt is presented, but $d\tau$ stays the same. If r is light-like instead, notice that $d\tau = 0$. Photons traveling in vacuum does not experience time in the same sense.

Time Dilation: Choose an inertial frame. Then r is parametrized as $r(t) = (ct, x(t), y(t), z(t))$. Let $v = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right)$. Then we have $d\tau = \sqrt{1 - |v|^2} dt$. So time experienced by an observer is proportional to proper time by a factor depending on the velocity of the observer.

The Twin Paradox: Among all world lines r passing through p and q , the geodesic one (the straight line) has the largest proper time. The twin paradox says that the twin taking the bendy road with acceleration ages faster than the other twin whose trajectory is straight. Mathematically, $r(\tau)$ is the unit speed parameterization of r .

Definition 1.2.3 (Unit Parametrization)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line. Define the unit parametrization of r by

$$\hat{r}(s) = (r \circ \tau^{-1})(s)$$

Definition 1.2.4 (Lorentz Factor)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line. Define the Lorentz factor of r to be

$$\gamma = \frac{1}{\sqrt{-\eta(r', r')}}.$$

Definition 1.2.5 (Four-Velocity and Four-Acceleration)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line.

- Define the four-velocity of r by

$$u(s_0) = \left. \frac{d\hat{r}}{ds} \right|_{s=s_0} = \gamma r'(s_0)$$

- Define the four-acceleration of r by

$$a(s_0) = \left. \frac{du}{ds} \right|_{s=s_0}$$

Because tangent spaces are canonically identified in Minkowski spacetime, we think of $a(s_0)$ as an element of the tangent space $T_{\gamma(s_0)}M$.

Proposition 1.2.6

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \rightarrow M$ be a world line. Then the following are true.

- $\eta(u, u) = -1$.
- $\eta(u, a) = 0$.

In Galilean spacetime, the three dimensional slices of the world along time is orthogonal to the flow of time. In Minkowski spacetime, the same is true, but time flows differently according to each observer (inertial frames).

Definition 1.2.7 (Simultaneity Hypersurface)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $v \in V$ be a future pointing time-like vector. Let $p \in M$. Define the simultaneity hypersurface through p relative to v by the space

$$\Sigma_p(v) = \{q \in M \mid \eta(q - p, v) = 0\}$$

A key point is that $\eta|_{\Sigma_p(v)}$ is positive definite, which means that it becomes an inner product on the three dimensional slice. Another key point: for the worldline r_1, r_2 of two observers at the same point p , their future pointing time-like vectors r'_1 and r'_2 are generally different, which gives a different three dimensional slice.

Length contraction: Suppose an metal rod exists in spacetime. The endpoints are given by two world lines r_1, r_2 such that $r_2 = q + r_1$ for some $q \in V$ and $\eta(r_2 - r_1, v)$ where here we choose an inertial frame to fix a future pointing vector v . Then define the length of the rod according to this observer to be the quantity

$$L = \sqrt{\eta(q, q)}$$

Then???

If a basis E is chosen for V such that

$$[\eta]_{E,E} = \begin{pmatrix} -c^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Then unit parameterization would instead be

$$\tau(t) = \frac{1}{c} \int_{t_0}^t \sqrt{-\eta(r'(s), r'(s))} ds$$

and we would have $\eta(u, u) = -c^2$. Time dilation is displayed as $d\tau = \sqrt{1 - |v|^2/c^2} dt$

1.3 Relativistic Dynamics

In classical mechanics, Galilean force laws are of the form

$$m \frac{d^2 r}{dt^2} = F$$

where F is invariant under Galilean transformations.

Definition 1.3.1 (Minkowski Force)

Let $(M, V, +, \eta)$ be Minkowski spacetime. A force law consists of the following data.

- A world line $r : I \subseteq \mathbb{R} \rightarrow M$.
- A constant $m \in \mathbb{R}$ called rest mass.
- A smooth vector field $F : I \rightarrow TM$ called a force such that

$$ma = F$$

Definition 1.3.2 (Four Momentum)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. Define the four momentum to be

$$p = mu$$

Proposition 1.3.3

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. Then the following are true.

- $\eta(p, p) = m^2$.
- $\eta(p, F) = 0$.

For the rest of the section here we assume that η is given specifically by the matrix that incorporates the speed of light c . Assume that ordinary Newtonian force law in three dimensions hold. In other words we assume that

$$\frac{d}{dt} \frac{mr'(t)}{\sqrt{1 - |r'(t)|^2/c^2}} = \mathbf{F}$$

for some Newtonian force law F (determined for example experimentally). Using $\eta(p, F) = 0$, we deduce that $F = \left(\frac{\gamma}{c} \mathbf{F} \cdot r'(t), \gamma \mathbf{F} \right)$. Comparing first components of a force law gives us

$$\frac{d}{dt} \frac{mc^2}{\sqrt{1 - |r'(t)|^2/c^2}} = \mathbf{F} \cdot r'(t)$$

Since integrating $\mathbf{F} \cdot r'(t)$ gives work done, we must make the following definition.

Definition 1.3.4 (Relativistic Energy)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. Define the relativistic energy to be

$$E = \gamma mc^2$$

Using the binomial theorem, we see that

$$E = mc^2 + \frac{1}{2}m|r'(t)|^2 + \dots$$

The higher terms contain velocity $|r'(t)|$. So the first term recovers Einstein's famous equation.

Proposition 1.3.5

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. Then we have

$$E^2 = p^2 c^2 + m^2 c^4$$

1.4 Coordinate Lagrangian Formulation

We first consider a single free particle in Minkowski spacetime. Ideally we want the Lagrangian to be invariant under relativistic quantities. So assume that

$$L = \alpha \gamma$$

for some $\alpha \in \mathbb{R}$. Here the Lorentz factor γ is thought to be the simplest relativistic invariant quantity. Then variational principle solution to the action $S = \int_a^b L dt$ is precisely the solution to the usual Euler Lagrange equation in classical mechanics. When there are other forces acting on the particle, we assume that it interacts on the Lagrangian simply.

Definition 1.4.1 (The Coordinate Lagrangian)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. A Lagrangian is a function $L : TM \times \mathbb{R} \rightarrow \mathbb{R}$ of the form

$$L = -\gamma mc^2 - V$$

for some function $V : TM \times \mathbb{R} \rightarrow \mathbb{R}$ called the potential.

The problem here is that V is in general not Lorentz invariant and so L is in general not Lorentz invariant.

Pros: Motions of equations are accurate and easy enough.

Cons: Not covariant inherently, although in the usual cases they are provable to be covariant.

1.5 Covariant Lagrangian Formulation

Pros: It is covariant. Cons: There are difficulties generalizing to N-body.

2 Electromagnetism in Special Relativity

2.1 Classical Electromagnetism

Coulomb's law states that for two point charges q_1, q_2 separated by a displacement r , the force of q_1 acting on q_2 is given by

$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|r|^2} r$$

and vice versa, where ϵ_0 is a constant called the vacuum permeability.

For a point charge q position at $x_0 \in \mathbb{R}^3$, the test charge at a general position $x \in \mathbb{R}^3$ feels the charge

$$E(x) = \frac{1}{4\pi\epsilon_0} \frac{q}{|x - x_0|^2} (x - x_0)$$

called the electromagnetic force. This is a smooth function $E : \mathbb{R}^3 \setminus \{x_0\} \rightarrow \mathbb{R}^3$. Then the force the test charge feels is given by

$$F = q_{\text{test}} E$$

The charge multiplied with the field.

Postulate: Superposition: If there are n point charges $q_1, \dots, q_n$ at positions $z_1, \dots, z_n$, then the test charge at a general position x feels the charge

$$E(x) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|x - z_i|^2} (x - z_i)$$

In other words the postulate assumes that the test charge with multiple point charges is summed.

Again, given n point charges $q_1, \dots, q_n$ at positions $z_1, \dots, z_n : \mathbb{R} \rightarrow \mathbb{R}^3$, define the charge density to be the function $\rho : \mathbb{R}^4 \rightarrow \mathbb{R}$ given by

$$\rho(x, t) = \sum_{i=1}^n q_i \delta^3(x - z_i)$$

where δ^3 is the 3D Dirac delta distribution, defined by $\int_{\mathbb{R}^3} f(x) \delta^3(x - z_i) dx = f(z_i)$.

We compute its integral over a region $\Omega \subseteq \mathbb{R}^3$ to be

$$\int_{\Omega} \rho(x, t) dx = \sum_{i=1}^n 1_{z_i \in \Omega} q_i$$

Gauss's law states that

$$\int_S E \cdot dA = \frac{1}{\epsilon} \sum_{i=1}^n 1_{z_i \in \Omega} q_i$$

for any region $S \subseteq \mathbb{R}^3$. Using charge density, we can rewrite the equation into

$$\nabla \cdot E = \frac{\rho}{\epsilon_0}$$

This is the first of Maxwell's equations.

The second of Maxwell's equations

$$\nabla \times E = 0$$

follows by a simple calculation.

It follows that there exists a scalar function $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}$ called the scalar potential such that

$$E = -\nabla \phi$$

Explicitly it is defined by

$$\phi(x) = - \int_{x_{\text{ref}}}^x E \cdot dl$$

where l is any path from x_{ref} to x .

Possion's equations follows from the above:

$$\nabla^2 \phi = - \frac{\rho}{\varepsilon_0}$$

When $\rho = 0$ so the region is charge free, this is Laplace's equation.

We turn to the magnetic side. A static charge only produces E . But a moving charge (a current) produces a magnetic field that influences objects via the magnetic force. Postulate: For a point charge q with position $x_0 : \mathbb{R} \rightarrow \mathbb{R}^3$, a magnetic field $B : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ is produced and is given by

$$B(x, t) = \frac{\mu_0}{4\pi} q r(t) \times \frac{x - x_0(t)}{|x - x_0(t)|^3}$$

Postulate: Superposition. when multiple moving point charge exist, the magnetic field is stacked additively. Taking the continuum limit gives

$$J = \rho(x, t) r'(t)$$

called the net charge flux density (current). r here means that velocity drift (the overall direction that the charges are moving roughly). In this case, the magnetic field is given by

$$B(x, t) = \frac{\mu_0}{4\pi} \int_{\mathbb{R}^3} J(x') \times \frac{x - x'}{|x - x'|^3} dx$$

Gauss's law the third of Maxwell's equations: $\nabla \cdot B = 0$ means that there are no magnetic monopoles. Ampere's law the last of Maxwell's equations:

$$\int_C B \, dl = \mu_0 I$$

where I is the total current through a surface S with boundary C .

2.2 The Relativistic Formulation

Definition 2.2.1 (Four Potentials)

Let $(M, V, +, \eta)$ be Minkowski spacetime. A four potential on M is a smooth vector field $A \in \mathfrak{X}(M)$.

Definition 2.2.2 (The Vector and Scalar Potential)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let A be a four potential.

- Define the scalar potential of A to be the first component $\phi = A^0$.
- Define the vector potential of A to be the remaining components (A^1, A^2, A^3) .

Recall that we may lower the indices from $(A^0, \dots, A^3)$ to $(A_0, \dots, A_3)$ using the musical isomorphism. Explicitly, we have

$$A_i = \sum_{j=0}^3 \eta_{ij} A^j$$

Definition 2.2.3 (The Electromagnetic Field Strength Tensor)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let A be a four potential. Define the electromagnetic field strength tensor to be the $(0, 2)$ -tensor F whose components at a chart $(U, x^1, \dots, x^4)$ is given

by

$$F_{ij} = \frac{\partial A_j}{\partial x^i} - \frac{\partial A_i}{\partial x^j}$$

If we think of A as a 1-form, then F really is just defined by $F = dA$. The fact that $dF = 0$ recovers the two source-free equations.

Definition 2.2.4 (The Levi-Civita Symbols)

Let $i, j, k \in \{1, 2, 3\}$. Define the Levi-Civita symbols by

$$\varepsilon_{ijk} = \begin{cases} 1 & \text{if } \begin{pmatrix} i & j & k \end{pmatrix} \text{ is an even permutation} \\ -1 & \text{if } \begin{pmatrix} i & j & k \end{pmatrix} \text{ is an odd permutation} \\ 0 & \text{otherwise} \end{cases}$$

Definition 2.2.5 (The Electric and Magnetic Field)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let A be a four potential. Choose an inertial frame of reference (M, t, x, y, z) .

- Define the electric field to be the smooth map $E : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ given by

$$E(t, x, y, z) = c(F_{01}, F_{02}, F_{03})(t, x, y, z)$$

- Define the magnetic field to be the smooth map $B : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ given by

$$B(t, x, y, z) = (-F_{23}, F_{13}, -F_{12})(t, x, y, z)$$

Next: how does boost transform these frame dependent objects.

Definition 2.2.6 (Four Current)

Let $(M, V, +, \eta)$ be Minkowski spacetime. A four current on M is a smooth vector field $J \in \mathfrak{X}(M)$.

Definition 2.2.7 (Charge Potential)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let J be a four current. Define the charge potential of J to be the first component $\rho = J^0$.

Source equation is determined by physical experiments.

Definition 2.2.8 (Source Equation)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let A be a four potential. Let J be a four current. A source equation is an equation of the form

$$\sum_{i=0}^3 \frac{\partial F^{ij}}{\partial x^i} = \mu_0 J^j$$

where $\mu_0 \in \mathbb{R}$ is a constant called the vacuum permeability.

This recovers the classical source equations.

2.3 The Relativistic Lagrangian in Electromagnetism

Definition 2.3.1 (Electromagnetic Force Law)

Let $(M, V, +, \eta)$ be Minkowski spacetime. An electromagnetic force law consists of the following data.

- A world line $r : I \subseteq \mathbb{R} \rightarrow M$.
- A constant $m \in \mathbb{R}$ called mass.
- A constant $q \in \mathbb{R}$ called charge.

- A four potential A .
- A smooth vector field $F : I \rightarrow TM$ whose components are given by

$$F^i = q \sum_{j=1}^n F_j^i u^j$$

such that (r, m, F) is a force law.

Definition 2.3.2 (The Electromagnetic Potential)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, q, A, F) be an electromagnetic force law. Define the electromagnetic potential to be the smooth function $V : TM \times \mathbb{R} \rightarrow \mathbb{R}$ given by

$$V = q\phi(r(t), t) - qr'(t) \cdot (A^2, A^3, A^4)(r(t), t)$$

Definition 2.3.3 (The Coordinate Lagrangian of Electromagnetism)

Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, q, A, F) be an electromagnetic force law. Define the coordinate Lagrangian to be

$$L = -\gamma mc^2 - V$$

Next: derive Euler Lagrange, check Lorentz-invariance.